

Manh-Hung Nguyen

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Research Interests

Dynamic Economic Theory; Climate and Energy Economics; Environmental Health; Innovation and AI Economics.

Positions

- **Jul 2008–present:** Researcher, Toulouse School of Economics (TSE-R, INRAE)
- **Jan 2011–Mar 2011:** Research Fellow, NUS Business School, Singapore
- **Jan 2009–Jun 2009:** Visiting Professor, Department of Economics, National University of Singapore
- **Jun 2007–Sep 2007:** Visiting Fellow, Department of Economics, Cornell University, USA

Education

- **PhD in Economics**, University of Paris 1 Panthéon-Sorbonne, 2007
- **MSc in Mathematics**, Hanoi Institute of Mathematics, 2003

Projects

1. **Leader** (09/2024–09/2025), *Climate Policy Uncertainty, Energy Transition and Firm Resilience*, Sophie Germain program (funded by the French Ministry for Europe and Foreign Affairs and the UK Department for Science, Innovation and Technology).
2. **Co-Leader**, *Socio-économiques du changement climatique au Viet Nam et stratégies d'adaptation (GEMMES VN)* (2020–2022).
3. **Co-Leader**, *AI Marketplace Project: A Business Model and Analysis* (Aus4Innovation Program – CSIRO, 2021–2023).
4. **Leader (WP6)**, *Socioeconomic Impacts, Vietnam Space Climate Observatory* (VietSCO project, SCO CES-Toulouse, 2021–2025).
5. **Co-Leader**, *Environmental Health and Climate Change* (AFD, 2019–2022).
6. **Member**, *Risk Analysis and Policy Assessment for Wheat Health (POLIRISK)*, Métaprogramme SMACH INRA (2016–2019).
7. **Leader**, *Impacts of Coastal Erosion in Vietnam: Theoretical Models and Empirical Approaches (VIET-EROSION)* (2017–2019).
8. **Leader**, *Assessment of Climate Change Impacts on Nghe An, Vietnam: Mitigation and Adaptation (VIETFLOOD)* (2012–2014).
9. **Key Member**, *Croissance Économique et Demande d'Énergie en présence du Progrès Technique Endogène (CEDEPTE)*, ANR/CNRS (2007–2009).

10. **Member**, *Insurance and Remote Sensing Data in Agriculture (CoEDITAg)*, ANR and INRAE (2023–2026).

Refereeing Activity

Econometrica; Economic Theory; European Economic Review; Journal of Public Economic Theory; Journal of Mathematical Economics; Macroeconomic Dynamics; Environmental Modeling & Assessment; Mathematical Social Sciences; Economic Analysis and Policy; Journal of Asian and African Studies; Ecological Modelling; Economic Bulletin.

Working Papers

1. **Epistemic Capital and Two-Trap Growth in the AI Era**, 2026, TSE Working Paper.
2. **Anneal–Neutral Estimation in Dynamic Discrete Choice Models: Geometry of Indifference and Neutral Smoothing**, 2026, Working Paper.
3. **Capital Sunk, Emissions Locked: The Economics of Energy Transitions under Carbon Constraints** (with Michel Moreaux and Jean-Pierre Amigues), 2026, TSE Working Paper.
4. **Climate Change, Labor Market Frictions and Inequality** (with A. Goenka, L. Liu, and H. Pang), 2025, TSE Working Paper.

Peer-Reviewed Publications

A. Dynamic Theory and Mathematical Economics

1. Goenka, A., Liu, L., & Nguyen, M.H. (2024). Modelling optimal lockdowns with waning immunity. *Economic Theory*, 77(1), 197–234.
2. Goenka, A., & Nguyen, M.H. (2020). General existence of competitive equilibrium in the growth model with an endogenous labor–leisure choice. *Journal of Mathematical Economics*, 91, 90–98.
3. Nguyen, M.H., & Nguyen-Van, P. (2016). Optimal endogenous growth with natural resources: Theory and evidence. *Macroeconomic Dynamics*, 20(8), 2173–2209.
4. Ha-Huy, T., Le Van, C., & Nguyen, M.H. (2016). Arbitrage and asset market equilibrium in infinite dimensional economies with short-selling and risk-averse expected utilities. *Mathematical Social Sciences*, 79, 30–39.
5. Goenka, A., Liu, L., & Nguyen, M.H. (2014). Infectious diseases and economic growth. *Journal of Mathematical Economics*, 50, 34–53.
6. Goenka, A., Le Van, C., & Nguyen, M.H. (2012). Existence of competitive equilibrium in an optimal growth model with heterogeneous agents and endogenous leisure. *Macroeconomic Dynamics*, 16, 33–51.
7. Le Van, C., Luong, T.B., Nguyen, M.H., & Nguyen, T.A. (2010). New technology, human capital, and growth in a developing country. *Mathematical Population Studies*, 17(4).
8. Ha-Huy, T., & Nguyen, M.H. (2010). No unbounded arbitrage, weak no market arbitrage and no arbitrage price system conditions: Equivalent conditions. *Journal of Mathematical Economics*, 46(1), 128–131.
9. Luu, D.V., & Nguyen, M.H. (2009). On alternative theorem and necessary optimality conditions for efficiency. *Optimization*, 58(1), 49–62.

10. Le Van, C., Nguyen, M.H., & Vailakis, Y. (2007). Equilibrium dynamics in an aggregative model of capital accumulation with heterogeneous agents and elastic labor. *Journal of Mathematical Economics*, 43, 287–317.
11. Nguyen, M.H., & Luu, D.V. (2006). On necessary conditions for efficiency in directionally differentiable multiobjective optimization problems. *Nonlinear Functional Analysis and Applications*, 12, 439–453.
12. Luu, D.V., & Nguyen, M.H. (2005). Invexity of constraint maps in mathematical programs. *Nonlinear Functional Analysis and Applications*, 9, 289–304.

B. Empirical Climate, Energy, and Environmental Health

1. Nguyen, M.H, Le,T.H, A Development of a New Measure for Renewable Energy Uncertainty in Vietnam by Using Natural Language Processing and Textual Analysis, *Energy Nexus*, forthcoming, 2026
2. Nguyen, M.H., Hoang, V.N., Nghiem, S., & Nguyen, L.A. (2025). The dynamic and heterogeneous effects of COVID-19 vaccination mandates in the USA. *Health Economics*, 34, 51–536.
3. Nguyen, C.V., Nguyen, M.H., & Nguyen, T.T. (2023). The impact of cold waves and heat waves on mortality: Evidence from a lower middle-income country. *Health Economics*, 32(6), 1220–1243.
4. Goenka, A., Liu, L., & Nguyen, M.H. (2021). COVID-19 and a green recovery? *Economic Modelling*, 104, 105639.
5. Aubert, C., Nguyen, M.H., & Reynaud, A. (2013). Living with floods: Flood protective behaviours and flood risk perception of Vietnam households. *The Geneva Papers on Risk and Insurance*, 38, 547–579.
6. Nguyen, L.A., Nguyen, M.H., Reynaud, A., & Simioni, M. (2024). A comparative study of residents and tourists' valuation for a heterogeneous environmental good: The case of coastal erosion. *Marine Policy*, 161, 106038.
7. Dang, H.A., Huynh, T.L.D., & Nguyen, M.H. (2023). Does the COVID-19 pandemic disproportionately affect the poor? Evidence from a six-country survey. *Journal of Economics and Development*.
8. Ha, L.T., Hung, N.M., & Van Chon, L. (2023). Decomposed and partial connectedness between economic globalization, non-renewable and renewable energy consumption in Vietnam. *Environmental Science and Pollution Research*, 30(55), 117061–117081.
9. Nguyen, L.A., Nguyen, M.H., Hoang, V.N., Reynaud, A., Simioni, M., & Wilson, C. (2023). Tourists' preferences and willingness to pay for protecting a World Heritage site from coastal erosion in Vietnam. *Environment, Development and Sustainability*, 1–22.
10. Luong, T.A., Nguyen, M.H., Truong, N.T.K., & Le, K. (2023). Rainfall variability and internal migration: The importance of agriculture linkage and gender inequality. *Economic Analysis and Policy*, 77, 326–336.
11. Nguyen, M.H., Le, C.V., & Atkinson, S.E. (2022). The production inefficiency of US electricity industry in the face of restructuring and emission reduction. *Journal of Economics and Development*, 24(4), 278–292.
12. Aubert, C., Dang, H.A., & Nguyen, M.H. (2022). The unequal impact of the COVID pandemic: Theory and evidence on health and economic outcomes for different income groups. *Revue française d'économie*, 43–77.
13. Reynaud, A., Nguyen, M.H., & Aubert, C. (2022). Correction to: Is there a demand for flood insurance in Vietnam? Results from a choice experiment. *Environmental Economics and Policy Studies*, 24(2), 305–305.

14. Le, M.H., Luong, C., & Nguyen, T.A. (2022). The relative roles of domestic and foreign capital in aggregate production of GDP and CO₂-equivalent emissions across OECD countries. *Environmental Science and Pollution Research*.
15. Nguyen, M.H., Le, D.P., & Vo, T.T. (2021). Vulnerability to natural disaster and welfare effect: A case study of flood risk in Vietnam's North Central region. *Journal of Asian and African Studies*, 56(8), 1879–1898.
16. Nguyen, M.H., Nguyen, T.L.A., Nguyen, T., Reynaud, A., Simioni, M., & Hoang, V.N. (2021). Economic analysis of choices among differing measures to manage coastal erosion in Hoi An (a UNESCO World Heritage Site). *Economic Analysis and Policy*, 70, 529–543.
17. Le, C., Nguyen, M.H., & Nguyen, L.A. (2020). The economic impact of climate change on the fishing industry in Vietnam. *Journal of Asian Economic and Business Research*.
18. Nguyen, M.H., Nguyen, L.A., & Nguyen, T. (2020). Economic valuation of the coastal erosion prevention program: A case study of Hoi An. *Journal of Asian Economic and Business Research*.
19. Le Van, Q., Viet Nguyen, T., & Nguyen, M.H. (2019). Sustainable development and environmental policy: The engagement of stakeholders in green products in Vietnam. *Business Strategy and the Environment*, 28(5), 675–687.
20. Nguyen, T.V., Nguyen, M.H., & Le Van, Q. (2018). Is green growth possible in Vietnam? The case of marine capture fisheries. *BioPhysical Economics and Resource Quality*, 3, 1–10.
21. Reynaud, A., Nguyen, M.H., & Aubert, C. (2018). Is there a demand for flood insurance in Vietnam? Results from a choice experiment. *Environmental Economics and Policy Studies*, 20, 593–617.
22. Le Van, Q., Nguyen, M.H., & Nguyen, T.V. (2018). Towards green growth in Vietnam: The case of marine capture fisheries. *Biophysical Economics and Resource Quality*, 3(9).
23. Reynaud, A., & Nguyen, M.H. (2016). Valuing flood risk reductions. *Environmental Modeling & Assessment*, 21(5), 603–617.
24. Minh, N.K., Nguyen, M.H., & Hung, N.V. (2011). Modeling transformation in Vietnam using smooth transitions. *Environmental Economics*, 2(3), 16–32.

Books, Reports, and Book Chapters

1. Espagne, E., Ngo-Duc, T., Nguyen, M.H., Pannier, E., Woillez, M.N., Drogoul, A., et al. (2021). *Climate Change in Viet Nam: Impacts and Adaptation. A COP26 Assessment Report of the GEMMES Viet Nam Project*. AFD, 600 pages.
2. *Climate Change in Vietnam: Impacts and Adaptation*. The Conversation.
3. *Environmental Health and Climate Change*, Chapter 3, GEMMES/AFD/COP26 (Lead author: M.-H. Nguyen et al.).
4. *The Resilience of Energy Systems under Climate Change*, Chapter 5, GEMMES/AFD/COP26 (Lead author: M.-H. Nguyen et al.).
5. *Choice Experiment Methods in Natural Disasters Management*, manuscript, Nafosted, 2021.

6. *Optimality Conditions and Optimal Economic Growth Models*. Academic Publishing, 2010. ISBN: 978-3-8383-6112-3.
7. *Selected Mathematics Problems: Methods and Comments*, 3rd ed. (in Vietnamese), Hanoi National University Publisher, 2004.

Teaching

- **Toulouse School of Economics**: Topics in Environmental Economics (M2-ERNA), 2015–present
- **National University of Singapore**: Advanced Mathematical Economics (PhD course EC6104), 2009
- **National University of Singapore**: Macroeconomic Analysis (EC4102), 2009